

Explica este cargo

Transaction-dispute intake for LATAM Bank: Spanish and Portuguese, for customers in Mexico, Colombia, and Argentina.

Team **Problem Crushers**, Factored AI & Data Hackathon 2026

- **Live demo:** <https://explain-this-charge.nicerock-692cf9dc.eastus2.azurecontainerapps.io>
- **Code:** github.com/mastersanto/factored-hackathon-2026-problem-crushers

All data is the organizers' synthetic dataset. Conversations, Portuguese cases, and the compliance-review list are team-generated and labelled.

1. The problem, from the bank's own data

- **Unrecognized and wrongful charges are 36.5% of all complaints:** 24,491 cases.
- They take a **median of 15-16 days** to resolve, and **about 20% miss their deadline**.
- Complaint contacts are resolved at first contact only **43.6%** of the time.
- **Satisfaction follows resolution, not speed:** CSAT is 3.0 when resolved at first contact and 2.0 when not.
- Scam calls pushing customers to "confirm a charge" make it worse. Outside the data, unrecognized transactions were 39.8% of Colombia's financial-regulator complaints in 2025.

Goal: resolve the doubt correctly on first contact, and when a person is needed, give them a complete, verified case.

2. The workflow: understand → decide → act → verify → escalate

The customer says...	The assistant...
"No reconozco un cargo de 1.250 en Uber"	finds it in <i>their own</i> records and explains merchant, amount, date, place, channel, and card, each sentence verified with its record
"Sí, fui yo"	closes the case, with no person needed
"No fui yo"	collects what happened, files one structured handoff , and states the customer's rights and legal deadline (Mexico, Colombia, Argentina), never a promised outcome
"Me llamaron del banco y me pidieron un código"	checks the bank's outbound record: real contact / no record / scam . If a code was shared, the case is urgent .
vague, out of scope, another customer's data, prompt injection	asks a question, abstains, or refuses

Required cases covered: **normal** (explain and close), **ambiguous** (clarify), **human-required** (claim, fraud, compliance hold).

3. Architecture: code decides, models interpret

- **A deterministic state machine** owns every step, tool, rule, and permission. **Claude Haiku 4.5** only interprets messages; **Claude Sonnet 5.5** only rewords verified facts.
- **Guarantees are enforced in code, not prompts:**
 - tools read only the **session's** customer;
 - a **faithfulness check** discards rewordings that change or add numbers, or promise anything;
 - a **close guard** means a model can never close a fraud claim on its own;
 - **compliance holds** are never explained, and the customer's device gets only a case number.
- **Rules mode:** on a model failure, a refusal, or a spent budget, the same guarantees hold without the model.
- **Data:** a minimized Parquet warehouse in DuckDB, with identity data dropped, card numbers cut to the last 4 digits, and 11 quality checks.
- **Learned component:** a calibrated fraud-risk estimate against the bank's fixed threshold, on a time split, tracked in MLflow.
- **Deployment:** one container on **Azure Container Apps**, with a private registry, the key as a secret, and scale to zero.

4. Results on held-out cases

180 cases per set: 15 categories × ES/PT, each tied to real records, with the expected outcome written before the run. The held-out set uses **phrasings never used for tuning**.

Held-out phrasings	Rules only	With Claude
Correct outcome	85.0%	95.6% (3 runs: 95.0–96.1)
Missed transfers (needed a person)	28.3%	2.2% (1.7–3.3)
Unnecessary transfers	0%	0%
Unsafe outcomes	0	0
p95 latency per turn · cost per case	0.12 s · \$0	3.7 s · \$0.0033

- **Baseline** (every case to an agent): 0% automation, a median 120 s wait plus 431 s of handling.
- **Fraud estimate: 281 of 494** test frauds caught at 100% precision, against **205** for the fixed rule (+37%).
- **The evaluation changed the system**: it found 3 fraud claims closed on a model misreading (now guarded), a Portuguese detection gap, and internal case data streamed to the browser. All three were fixed, with tests.

5. Limits, and what production needs

- **Synthetic data:** several fields are random, there is no MXN, and only 24 merchants. The fraud score may encode the label. Conversations and Portuguese are team-generated.
- **Rules from desk research:** the financial specialist's validation is pending.
- **Production needs:**
 - the bank's identity service, and authentication for the specialist queue;
 - live read-only services behind the same tools;
 - a retention policy and legal review of the wording, in each country;
 - several replicas and a handoff queue;
 - monitoring of the unsafe-outcome checks, fallback rate, latency, and cost;
 - fairness and model-risk review.

Details: `docs/limitations.md` , `docs/evaluation.md` , and `docs/model-card.md` in the repository.